

lab24(SSU)

December 1, 2025

1 Lab 24: World Population and Poverty

In this project, you'll explore data from Gapminder.org, a website dedicated to providing a fact-based view of the world and how it has changed. That site includes several data visualizations and presentations, but also publishes the raw data that we will use in this project to recreate and extend some of their most famous visualizations.

The Gapminder website collects data from many sources and compiles them into tables that describe many countries around the world. All of the data they aggregate are published in the [Systema Globalis](https://SystemaGlobalis.org). Their goal is “to compile all public statistics; Social, Economic and Environmental; into a comparable total dataset.” All data sets in this project are copied directly from the Systema Globalis without any changes.

This project is dedicated to [Hans Rosling](https://www.hansrosling.se/) (1948-2017), who championed the use of data to understand and prioritize global development challenges.

```
[1]: !apt-get install texlive texlive-xetex texlive-latex-extra pandoc
      !pip install py pandoc
      !pip install datascience
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
pandoc is already the newest version (2.9.2.1-3ubuntu2).
pandoc set to manually installed.
The following additional packages will be installed:
  dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
  fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
  libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
  libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
  libsynchronet2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzzip-0-13
  lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
  ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
  teckit tex-common tex-gyre texlive-base texlive-binaries
  texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
  texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils
Suggested packages:
  fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
  libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
```

```

poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc

```

The following NEW packages will be installed:

```

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils

```

0 upgraded, 54 newly installed, 0 to remove and 41 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-droid-fallback all 1:6.0.1r16-1.1build1 [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-lato all 2.0-2.1 [2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 poppler-data all 0.4.11-1 [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-common all 6.17 [33.7 kB]

Get:5 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-urw-base35 all 20200910-1 [6,367 kB]

Get:6 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9-common all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]

Get:7 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libidn12 amd64 1.38-4ubuntu1 [60.0 kB]

Get:8 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libijs-0.35 amd64 0.35-15build2 [16.5 kB]

Get:9 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libjbig2dec0 amd64 0.19-3build2 [64.7 kB]

Get:10 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9 amd64 9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]

Get:11 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libkpathsea6 amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]

Get:12 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libwoff1 amd64 1.0.2-1build4 [45.2 kB]

Get:13 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
Get:14 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-lmodern all
2.004.5-6.1 [4,532 kB]
Get:15 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all
20201225-1build1 [397 kB]
Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all
20180621-3.1 [10.2 MB]
Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java
all 18-1 [4,720 B]
Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-
java all 43-1 [10.8 kB]
Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-
java all 1.2-2 [60.3 kB]
Get:20 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1
amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration
all 1.18 [5,336 B]
Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64
3.0.2-7ubuntu2.11 [50.1 kB]
Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems
all 3.3.5-2ubuntu1.2 [228 kB]
Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1
[5,100 B]
Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7
kB]
Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all
0.1.1-2 [12.6 kB]
Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick
all 1.7.0-3ubuntu0.2 [52.5 kB]
Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all
0.3.2-1ubuntu0.1 [24.9 kB]
Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0
amd64 3.0.2-7ubuntu2.11 [5,114 kB]
Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsynchronet2
amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
Get:31 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64
2.5.11+ds1-1 [421 kB]
Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53
amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
Get:33 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2
amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
Get:34 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64
0.13.72+dfsg.1-1.1 [27.0 kB]
Get:35 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all
1:1.0.5-0ubuntu2 [578 kB]
Get:36 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-utils amd64
1:7.7+6build2 [94.6 kB]

```

Get:37 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:38 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
Get:39 http://archive.ubuntu.com/ubuntu jammy/main amd64 tiutils amd64
1.41-4build2 [61.3 kB]
Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:42 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all
2021.20220204-1 [14.3 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 7s (27.2 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 121713 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.

```

```

Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...

```

```

Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluaajit2:amd64.
Preparing to unpack
.../32-libtexluaajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...

```

```

Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libbzip-0-13:amd64.
Preparing to unpack .../33-libbzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libbzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../45-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../46-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../47-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.

```

```

Preparing to unpack .../48-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../49-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../50-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../51-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../52-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../53-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0-dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0-dfsg1-0ubuntu5.13) ...

```



```

Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmulubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

```

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a symbolic link

Processing triggers for tex-common (6.17) ...

Running updmap-sys. This may take some time... done.

Running mktexlsr /var/lib/texmf ... done.

Building format(s) --all.

This may take some time... done.

Collecting py pandoc

Downloading py pandoc-1.16.2-py3-none-any.whl.metadata (16 kB)

Downloading py pandoc-1.16.2-py3-none-any.whl (19 kB)

Installing collected packages: py pandoc

Successfully installed py pandoc-1.16.2

Collecting datascience

Downloading datascience-0.18.1-py3-none-any.whl.metadata (910 bytes)

Requirement already satisfied: folium>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from datascience) (0.20.0)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from datascience) (75.2.0)

Requirement already satisfied: matplotlib>=3.0.0 in

/usr/local/lib/python3.12/dist-packages (from datascience) (3.10.0)
 Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
 (from datascience) (2.2.2)
 Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
 (from datascience) (1.16.3)
 Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
 (from datascience) (2.0.2)
 Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-
 packages (from datascience) (7.34.0)
 Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages
 (from datascience) (5.24.1)
 Requirement already satisfied: branca in /usr/local/lib/python3.12/dist-packages
 (from datascience) (0.8.2)
 Requirement already satisfied: jinja2>=2.9 in /usr/local/lib/python3.12/dist-
 packages (from folium>=0.9.1->datascience) (3.1.6)
 Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-
 packages (from folium>=0.9.1->datascience) (2.32.4)
 Requirement already satisfied: xyzservices in /usr/local/lib/python3.12/dist-
 packages (from folium>=0.9.1->datascience) (2025.11.0)
 Requirement already satisfied: contourpy>=1.0.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (1.3.3)
 Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-
 packages (from matplotlib>=3.0.0->datascience) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (4.60.1)
 Requirement already satisfied: kiwisolver>=1.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (1.4.9)
 Requirement already satisfied: packaging>=20.0 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (25.0)
 Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
 packages (from matplotlib>=3.0.0->datascience) (11.3.0)
 Requirement already satisfied: pyparsing>=2.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (3.2.5)
 Requirement already satisfied: python-dateutil>=2.7 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (2.9.0.post0)
 Collecting jedi>=0.16 (from ipython->datascience)
 Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)
 Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (4.4.2)
 Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (0.7.5)
 Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.12/dist-

packages (from ipython->datascience) (5.7.1)
 Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in
 /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (3.0.52)
 Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (2.19.2)
 Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (0.2.0)
 Requirement already satisfied: matplotlib-inline in
 /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.2.1)
 Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (4.9.0)
 Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
 packages (from pandas->datascience) (2025.2)
 Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
 packages (from pandas->datascience) (2025.2)
 Requirement already satisfied: tenacity>=6.2.0 in
 /usr/local/lib/python3.12/dist-packages (from plotly->datascience) (9.1.2)
 Requirement already satisfied: parso<0.9.0,>=0.8.4 in
 /usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->datascience)
 (0.8.5)
 Requirement already satisfied: MarkupSafe>=2.0 in
 /usr/local/lib/python3.12/dist-packages (from
 jinja2>=2.9->folium>=0.9.1->datascience) (3.0.3)
 Requirement already satisfied: ptyprocess>=0.5 in
 /usr/local/lib/python3.12/dist-packages (from pexpect>4.3->ipython->datascience)
 (0.7.0)
 Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
 packages (from prompt-
 toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0->ipython->datascience) (0.2.14)
 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
 packages (from python-dateutil>=2.7->matplotlib>=3.0.0->datascience) (1.17.0)
 Requirement already satisfied: charset_normalizer<4,>=2 in
 /usr/local/lib/python3.12/dist-packages (from
 requests->folium>=0.9.1->datascience) (3.4.4)
 Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
 packages (from requests->folium>=0.9.1->datascience) (3.11)
 Requirement already satisfied: urllib3<3,>=1.21.1 in
 /usr/local/lib/python3.12/dist-packages (from
 requests->folium>=0.9.1->datascience) (2.5.0)
 Requirement already satisfied: certifi>=2017.4.17 in
 /usr/local/lib/python3.12/dist-packages (from
 requests->folium>=0.9.1->datascience) (2025.11.12)
 Downloading datascience-0.18.1-py3-none-any.whl (725 kB)
 725.1/725.1 kB
 12.8 MB/s eta 0:00:00
 Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
 1.6/1.6 MB
 47.3 MB/s eta 0:00:00

Installing collected packages: jedi, datascience
Successfully installed datascience-0.18.1 jedi-0.19.2

```
[2]: # Connect Google Drive to Colab so you can access your files
from google.colab import drive
drive.mount('/content/drive')

import os
os.chdir('/content/drive/MyDrive/Colab Notebooks/')
```

Mounted at /content/drive

```
[3]: # Run this cell to set up the notebook, but please don't change it.

# These lines import the NumPy and Datascience modules.
from datascience import *
import numpy as np

# These lines do some fancy plotting magic.
%matplotlib inline
import matplotlib.pyplot as plots
plots.style.use('fivethirtyeight')

from ipywidgets import interact, interactive, fixed, interact_manual
import ipywidgets as widgets
```

1.1 1. Global Population Growth

The global population of humans reached 1 billion around 1800, 3 billion around 1960, and 7 billion around 2011. The potential impact of population growth has concerned scientists, economists, and politicians alike.

The United Nations Population Division estimates that the world population will likely continue to grow throughout the 21st century, but at a slower rate, perhaps reaching and stabilizing at 11 billion by 2100. However, the UN does not rule out scenarios of slower or more extreme growth. These projections help us understand long-term population processes, even if they leave out possible global catastrophic events like war or climate crises.

In this part of the project, we will examine some of the factors that influence population growth and how they have been changing over the years and around the world. There are two main sub-parts of this analysis.

- First, we will examine the data for one country, Poland. We will see how factors such as life expectancy, fertility rate, and child mortality have changed over time in Poland, and how they are related to the rate of population growth.
- Next, we will examine whether the changes we have observed for Poland are particular to that country or whether they reflect general patterns observable in other countries too. We will study aspects of world population growth and see how they have been changing.

The first table we will consider contains the total population of each country over time. Run the cell below.

```
[4]: population = Table.read_table('./DS/population.csv').where("time", are.  
    ↳ below(2021))  
population.show(3)
```

<IPython.core.display.HTML object>

Note: The population csv file can also be found [here](#).

1.1.1 Poland

The Central European nation of Poland has undergone many changes over the centuries. In modern times it was (re)created as a democratic republic in 1919 after World War I. It was invaded and divided in World War II between Germany and the Soviet Union. War and the Holocaust had a devastating impact on its people. Poland was constituted in its current borders at the end of World War II (1945) under a communist government. In 1989, with the fall of the Soviet Union, Poland re-established itself as a democratic republic.

In this section of the project, we will examine aspects of the population of Poland since 1900. Poland's borders have changed, so we will look at the population within its current (2012) borders.

In the `population` table, the `geo` column contains three-letter codes established by the [International Organization for Standardization](#) (ISO) in the [Alpha-3](#) standard. **Use the [Alpha-3](#) link to find the 3-letter code for Poland.**

Question 1. Create a table called `p_pop` that has two columns labeled `time` and `population_total`. The first column should contain the years from 1900 through 2020 (including both 1900 and 2020) and the second should contain the population of Poland in each of those years.

```
[35]: p_pop = population.where('geo', 'pol').where('time', are.  
    ↳ between_or_equal_to(1900, 2020)).select('time', 'population_total')  
p_pop
```

```
[35]: time | population_total  
1900 | 24340168  
1901 | 24517983  
1902 | 24694172  
1903 | 24868705  
1904 | 25044468  
1905 | 25221471  
1906 | 25399722  
1907 | 25579229  
1908 | 25760003  
1909 | 25848341  
... (111 rows omitted)
```

Run the following cell to create a table called `p_five` that has the population of Poland every five years.

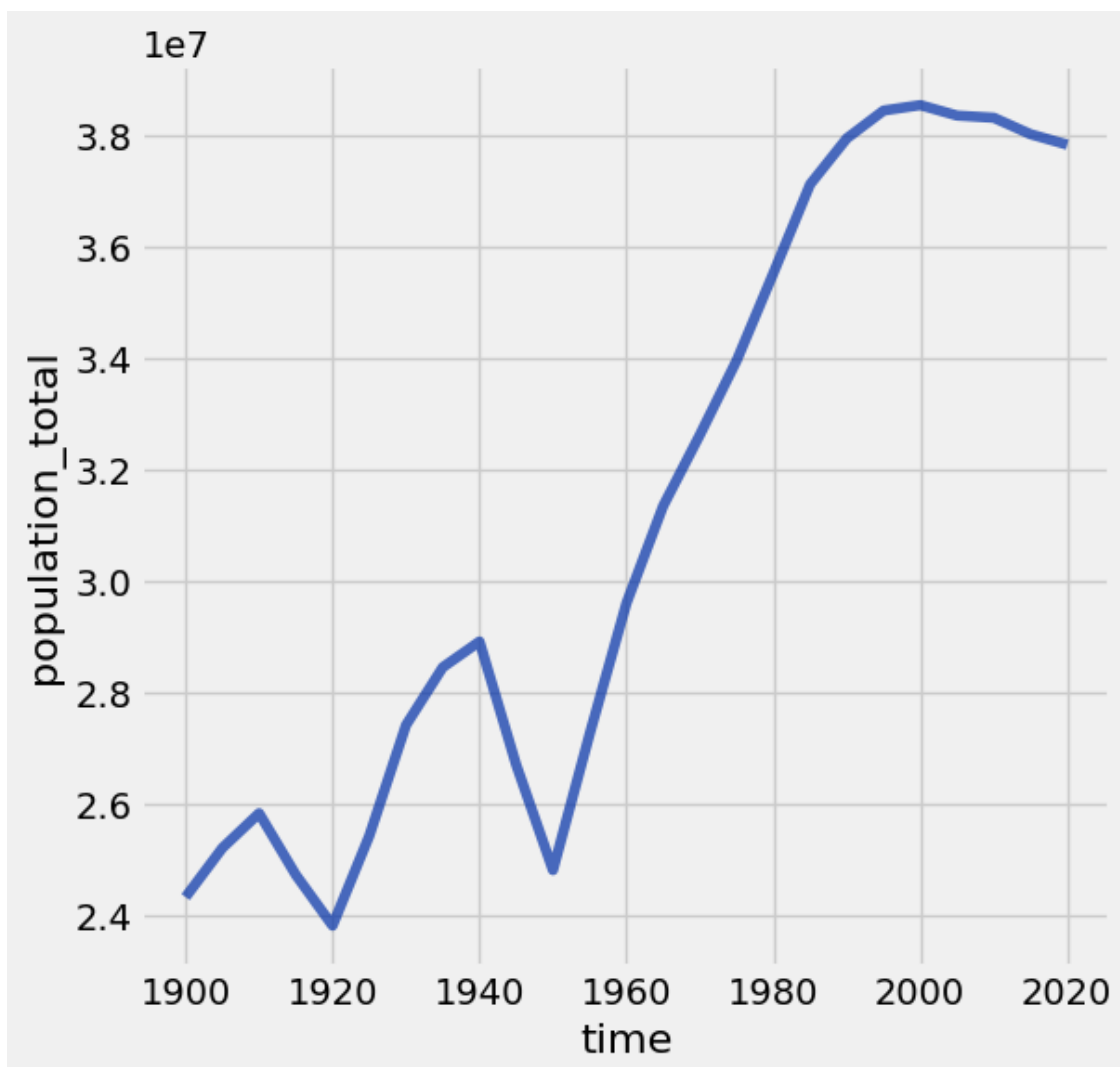
```
[36]: p_pop.set_format('population_total', NumberFormatter)

fives = np.arange(1900, 2021, 5) # 1900, 1905, 1910, ...
p_five = p_pop.sort('time').where('time', are.contained_in(fives))
p_five.show(3)
```

<IPython.core.display.HTML object>

Run the following cell to visualize the population over time. Following the devastating effects of World War I and World War II, Poland's population increased steadily from 1950 to 2000 and then leveled off. In the following questions we'll investigate this period of population growth.

```
[37]: p_five.plot(0, 1)
```



Question 2. Assign `initial` to an array that contains the population for every five year interval from 1900 to 2015 (inclusive). Then, assign `changed` to an array that contains the population for

every five year interval from **1905 to 2020** (inclusive). The first array should include both 1900 and 2015, and the second array should include both 1905 and 2020. You should use the `p_five` table to create both arrays, by first filtering the table to only contain the relevant years.

The annual growth rate for a time period is equal to:

$$\left(\left(\frac{\text{Population at end of period}}{\text{Population at start of period}} \right)^{\frac{1}{\text{number of years}}} \right) - 1$$

We have provided the code below that uses `initial` and `changed` in order to add a column to `p_five` called `annual_growth`. **Don't worry about the calculation of the growth rates**; run the test below to test your solution.

If you are interested in how we came up with the formula for growth rates, consult the [growth rates](#) section of the textbook.

```
[38]: initial = p_five.where('time', are.between_or_equal_to(1900, 2015)).
      ↪column('population_total')
      changed = p_five.where('time', are.between_or_equal_to(1905, 2020)).
      ↪column('population_total')

      p_1900_through_2015 = p_five.where('time', are.below_or_equal_to(2015))
      p_five_growth = p_1900_through_2015.with_column('annual_growth', (changed/
      ↪initial)**0.2-1)
      p_five_growth.set_format('annual_growth', PercentFormatter)
```

```
[38]: time | population_total | annual_growth
      1900 | 24,340,168      | 0.71%
      1905 | 25,221,471      | 0.49%
      1910 | 25,844,639      | -0.87%
      1915 | 24,739,041      | -0.75%
      1920 | 23,827,906      | 1.32%
      1925 | 25,439,942      | 1.52%
      1930 | 27,427,399      | 0.75%
      1935 | 28,465,230      | 0.32%
      1940 | 28,927,566      | -1.58%
      1945 | 26,713,772      | -1.46%
      ... (14 rows omitted)
```

The annual growth rate in Poland has been declining since 1950, as shown in the table below.

```
[39]: # Run this cell to view annual growth rates in Poland since 1950.
      p_five_growth.where('time', are.above_or_equal_to(1950)).show()
```

<IPython.core.display.HTML object>

Next, we'll try to understand what has changed in Poland that might explain the slowing population growth rate. Run the next cell to load three additional tables of measurements about countries

over time.

```
[40]: life_expectancy = Table.read_table('./DS/life_expectancy.csv').where('time', are.
      ↪are.below(2021))
      child_mortality = Table.read_table('./DS/child_mortality.csv').relabel(2,
      ↪'child_mortality_under_5_per_1000_born').where('time', are.below(2021))
      fertility = Table.read_table('./DS/fertility.csv').where('time', are.
      ↪below(2021))
```

The `life_expectancy` table contains a statistic that is often used to measure how long people live, called *life expectancy at birth*. This number, for a country in a given year, **does not measure how long babies born in that year are expected to live**. Instead, it measures how long someone would live, on average, if the *mortality conditions* in that year persisted throughout their lifetime. These “mortality conditions” describe what fraction of people for each age survived the year. So, it is a way of measuring the proportion of people that are staying alive, aggregated over different age groups in the population.

Run the following cells below to see `life_expectancy`, `child_mortality`, and `fertility`. Refer back to these tables as they will be helpful for answering further questions!

```
[41]: life_expectancy.show(3)
```

<IPython.core.display.HTML object>

```
[42]: child_mortality.show(3)
```

<IPython.core.display.HTML object>

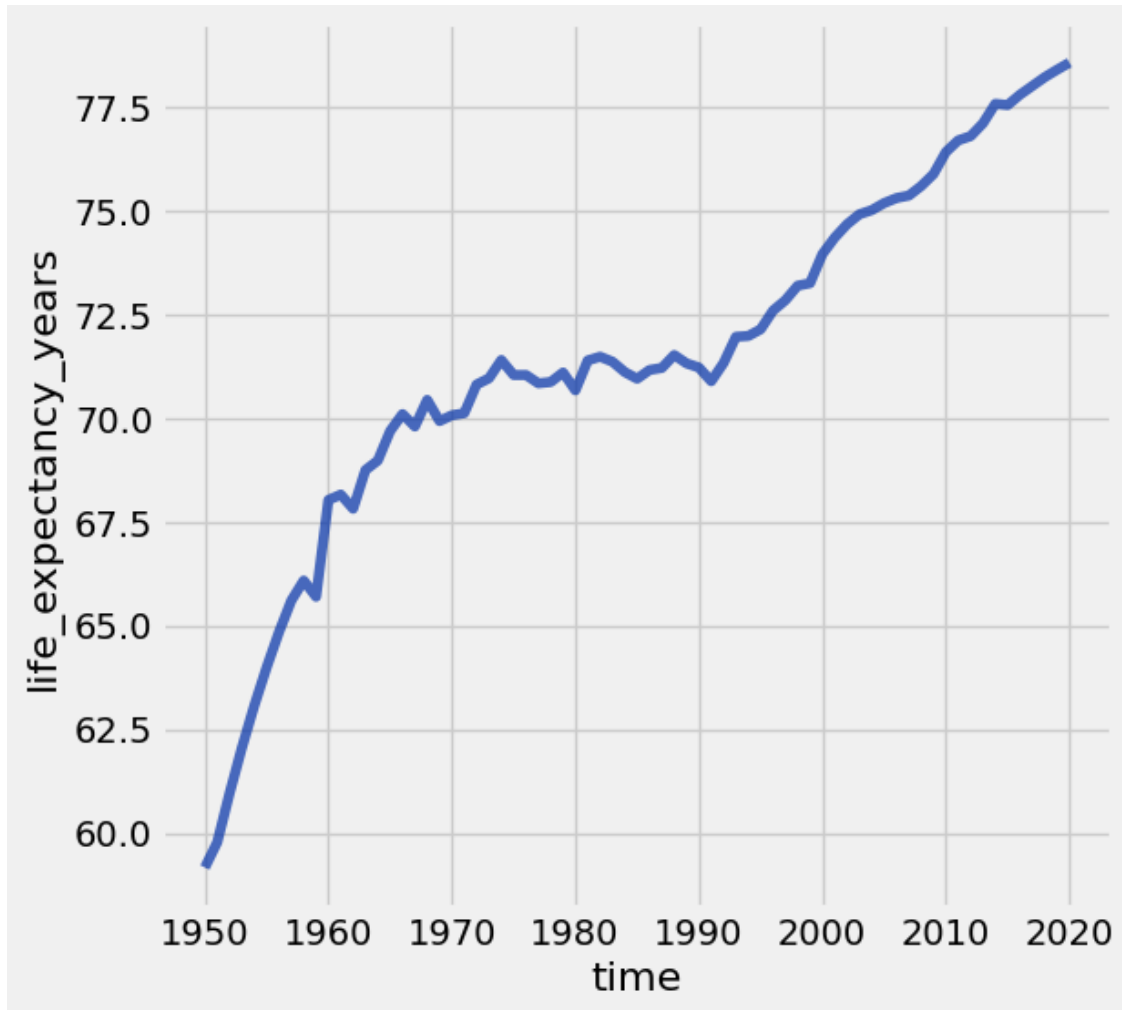
```
[43]: fertility.show(3)
```

<IPython.core.display.HTML object>

Question 3. Is population growing more slowly perhaps because people aren’t living as long? Use the `life_expectancy` table to draw a line graph with the years 1950 and later on the horizontal axis that shows how the *life expectancy at birth* has changed in Poland.

Hint: Make sure you filter the table properly; otherwise, the graph may look funky!

```
[44]: # Fill in code here
      poland_life_expectancy = life_expectancy.where('geo', 'pol').where('time', are.
      ↪above_or_equal_to(1950))
      poland_life_expectancy.plot('time', 'life_expectancy_years')
```



Question 4. Assuming everything else stays the same, do the trends in life expectancy in the graph above directly explain why the population growth rate decreased since 1950 in Poland? Why or why not?

As the graph shows, life expectancy in Poland has steadily increased since 1950. All other things being equal, increased life expectancy contributes to population growth and is unlikely to be a direct cause of a decline in population growth. Therefore, the trend of increased life expectancy does not directly explain Poland's decline in population growth since 1950. Rather, other factors are likely at play.

The `fertility` table contains a statistic that is often used to measure how many babies are being born, the *total fertility rate*. This number describes the [number of children a woman would have in her lifetime](#), on average, if the current rates of birth by age of the mother persisted throughout her child bearing years, assuming she survived through age 49.

Question 5. Complete the function `fertility_over_time`. It takes two input arguments, the Alpha-3 code of a country (denoted as `country_code`) and a year to start from (denoted as `start`). It returns a two-column table with the column labels `Year` and `Children per woman`. These

columns can be used to generate a line chart of the country's fertility rate each year, starting from the year given by `start`. The plot should include the start year and all later years that appear in the fertility table.

Then, determine the Alpha-3 code for **Poland**. The code at the very bottom for `poland_code` and the year 1950 are inputted to your `fertility_over_time` function. The function returns a table which we use in order to plot how Poland's fertility rate has changed since 1950. Note that the function `fertility_over_time` should not return the plot itself – it returns a two column table. The expression that draws the line plot is provided for you; please don't change it.

Hint: Read about `tbl.relabeled` in the [Python Reference](#) to rename columns.

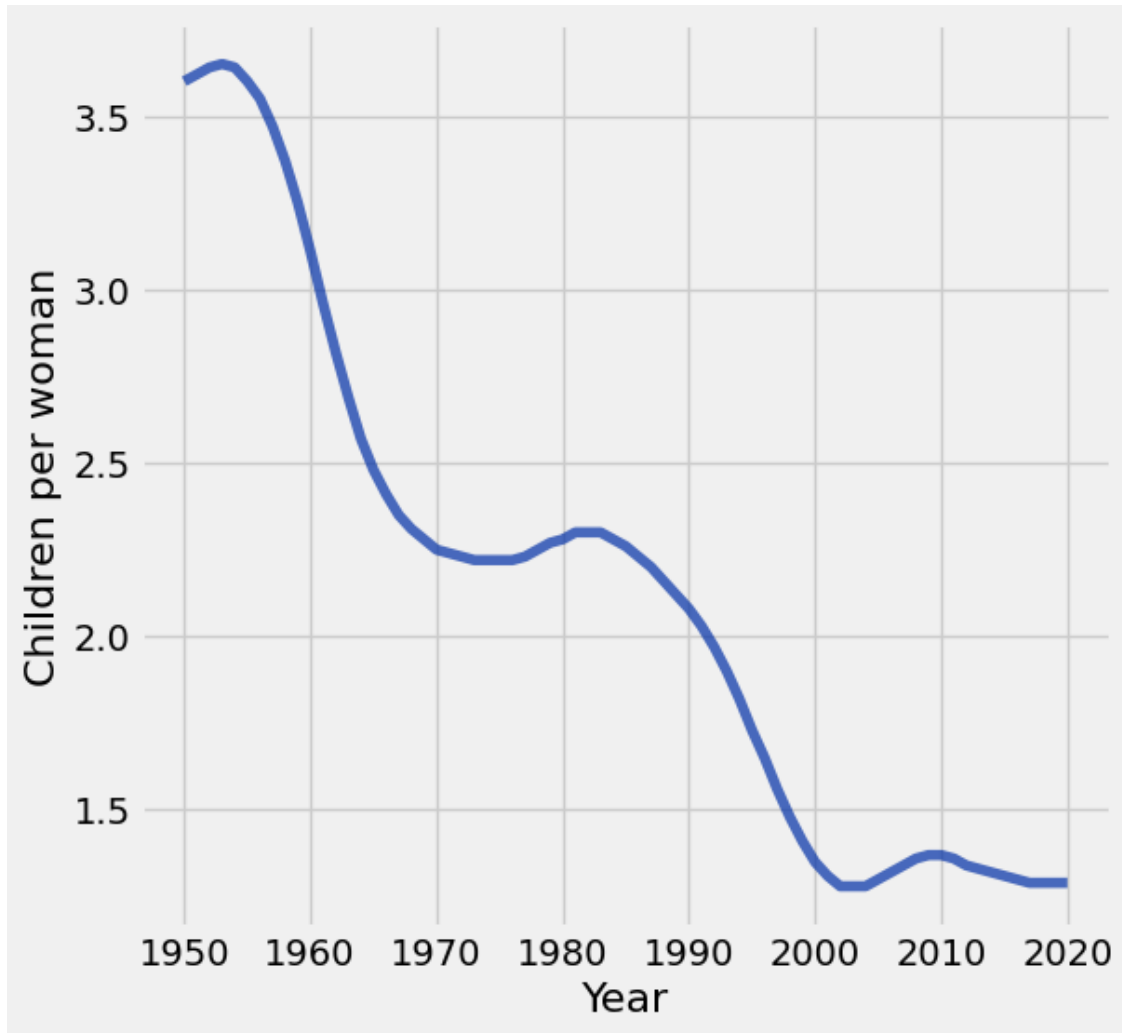
```
[52]: def fertility_over_time(country_code, start):
        """Create a two-column table that describes a country's total fertility
        ↪rate each year."""
        # It's a good idea (but not required) to use multiple lines in your
        ↪solution.
        return fertility.where('geo', country_code).where('time', are.
        ↪above_or_equal_to(start)).select('time',
        ↪'children_per_woman_total_fertility').relabeled('time', 'Year').
        ↪relabeled('children_per_woman_total_fertility', 'Children per woman')

        poland_code = 'pol'
        fertility_over_time(poland_code, 1950)
```

```
[52]: Year | Children per woman
      1950 | 3.6
      1951 | 3.62
      1952 | 3.64
      1953 | 3.65
      1954 | 3.64
      1955 | 3.6
      1956 | 3.55
      1957 | 3.47
      1958 | 3.37
      1959 | 3.25
      ... (61 rows omitted)
```

Plotting the fertility rate in Poland since 1950, we see a downward trend.

```
[53]: fertility_over_time(poland_code, 1950).plot(0, 1)
```



Question 6. Assuming everything else is constant, do the trends in fertility in the graph above help directly explain why the population growth rate decreased from 1950 to 2020 in Poland? Why or why not?

Looking at the graph in question 5, Poland's fertility rate (the number of children per woman) shows a consistent trend from 1950 to 2020. Assuming all other conditions hold, a declining birth rate directly means fewer babies are born. This is the main reason why the declining rate is continuing. This helps directly explain the projections from 1950 to 2020.

It has been [observed](#) that lower fertility rates are often associated with lower child mortality rates. We can see if this association is evident in Poland by plotting the relationship between total fertility rate and [child mortality rate per 1000 children](#).

Question 7. Create a table `poland_since_1950` that contains one row per year starting with 1950 and:

- A column `Year` containing the year
- A column `Children per woman` describing total fertility in Poland that year

- A column Child deaths per 1000 born describing child mortality in Poland that year

```
[55]: pol_fertility = fertility_over_time(poland_code, 1950) # Try starting with the
      ↪ table you built already!
      # It's a good idea (but not required) to use multiple lines in your solution.
      pol_child_mortality = child_mortality.where('geo', 'pol').where('time', are.
      ↪ above_or_equal_to(1950)).select('time',
      ↪ 'child_mortality_under_5_per_1000_born').relabelled('time', 'Year').
      ↪ relabeled('child_mortality_under_5_per_1000_born', 'Child deaths per 1000
      ↪ born')
      pol_fertility_and_child_mortality = pol_fertility.join('Year',
      ↪ pol_child_mortality)
      poland_since_1950 = pol_fertility_and_child_mortality
      poland_since_1950
```

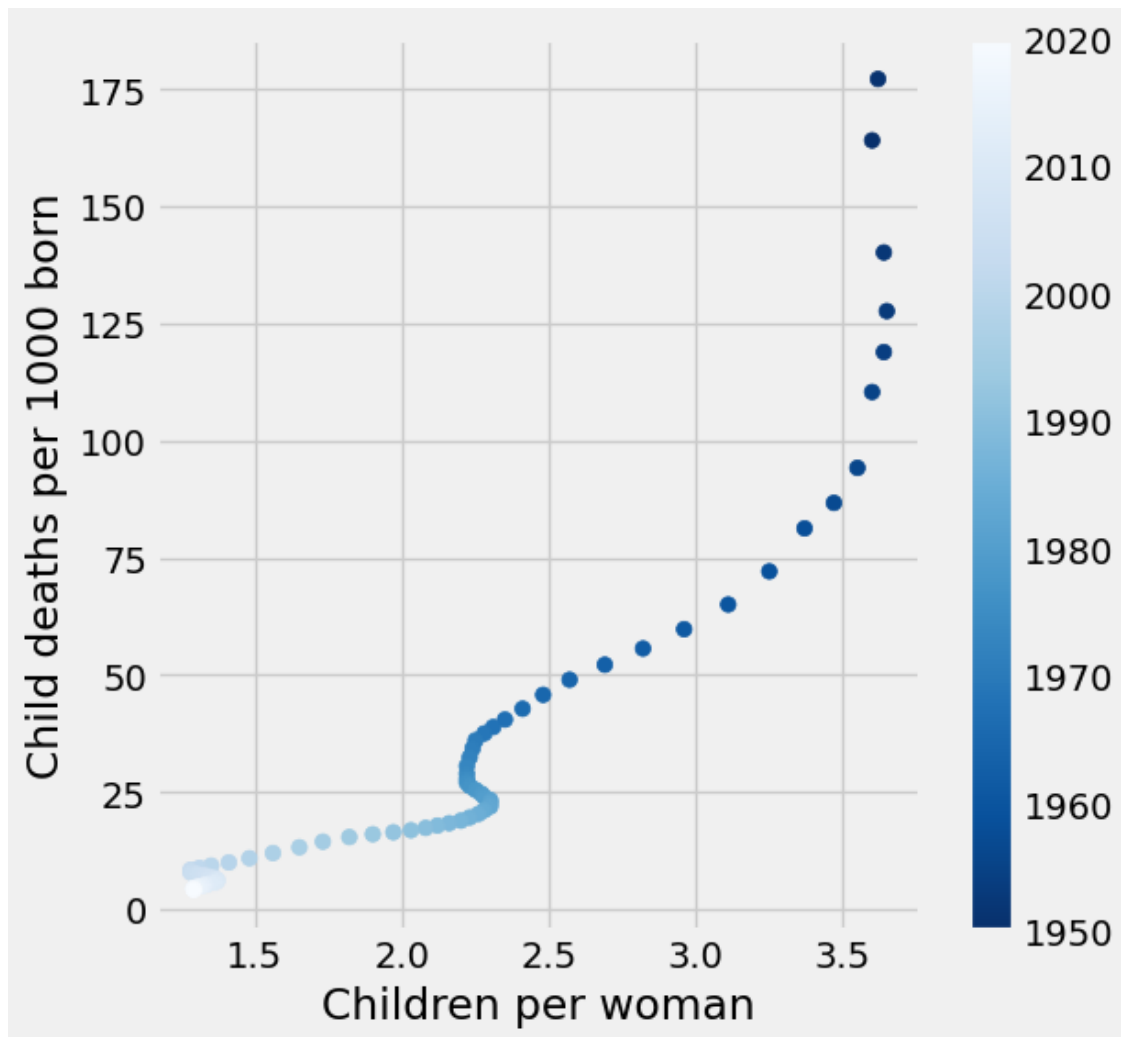
```
[55]: Year | Children per woman | Child deaths per 1000 born
      1950 | 3.6                | 164.08
      1951 | 3.62               | 177.18
      1952 | 3.64               | 140.16
      1953 | 3.65               | 127.65
      1954 | 3.64               | 118.9
      1955 | 3.6                | 110.38
      1956 | 3.55               | 94.19
      1957 | 3.47               | 86.72
      1958 | 3.37               | 81.28
      1959 | 3.25               | 72.1
      ... (61 rows omitted)
```

Run the following cell to generate a scatter plot from the `poland_since_1950` table you created.

The plot uses **color** to encode data about the **Year** column. The colors, ranging from dark blue to white, represent the passing of time between 1950 and 2020. For example, a point on the scatter plot representing data from the 1950s would appear as **dark blue** and a point from the 2010s would appear as **light blue**.

```
[56]: x_births = poland_since_1950.column("Children per woman")
      y_deaths = poland_since_1950.column("Child deaths per 1000 born")
      time_colors = poland_since_1950.column("Year")

      plots.figure(figsize=(6,6))
      plots.scatter(x_births, y_deaths, c=time_colors, cmap="Blues_r")
      plots.colorbar()
      plots.xlabel("Children per woman")
      plots.ylabel("Child deaths per 1000 born");
```



Question 8. In one or two sentences, describe the association (if any) that is illustrated by this scatter plot. Does the diagram show any causal relation between fertility and child mortality?

This scatterplot demonstrates a strong negative correlation between “births per woman” and “infant mortality rate per 1,000 newborns” in Poland. In other words, as the number of births decreases over time, so does the infant mortality rate. The points on the graph move diagonally from upper right to lower left, shifting from dark blue (past) to light blue (recent).

However, this diagram alone does not conclusively demonstrate a direct causal relationship between fertility and infant mortality. While the two variables show a strong correlation, it is more likely that multiple complex social and economic factors, such as economic development, increased access to healthcare, rising education levels, and improved sanitation, are simultaneously influencing both variables. For example, a lower infant mortality rate might lead parents to have fewer children, but this is more of an indirect interaction than a direct causal relationship. While this graph visually demonstrates the link between these trends, it is difficult to say that one variable directly “causes” the other.

Optional food for thought: What other context or information you would need in order to better understand the factors affecting life expectancy, child mortality, and fertility?

1.2 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit. **Please save before exporting!**

```
[ ]: # should change the directory and file name matching to yours
!jupyter nbconvert './DS/lab24(SSU).ipynb' --to pdf
```